SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID – SIH25060**
- **Problem Statement Title- Real Life Solutions for Waste Management**
- **Theme- Clean & Green Technology**
- **PS Category- Software**
- **Team ID-**
- **Team Name (Registered on portal): Buddy Bytes**

Buddy Bytes

Swachh AI: AI-Powered Waste Management Platform

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Idea/solutions:

- Automates waste type detection using **Google Gemini AI** for accurate reporting.
- Incentivizes citizens through a **gamified reward system** with redeemable points.
- Optimizes pickups via **GPS-enabled real-time collection tracking** and task assignment.
- Boosts participation with **community leaderboards** for areas and organizations.
- Educates users using **AR-based segregation guides** and in-app learning modules.
- Empowers ULBs with a **live dashboard, heatmaps, and predictive analytics**.

CHALLENGES ADDRESSED:

- Improves waste collection and treatment efficiency.
- Reduces unsegregated and unaccounted waste.
- Boosts citizen participation in waste management.
- Enables real-time monitoring and reporting.

KEY FEATURES:

- **AI Waste Detection** – Automatically identifies waste.
- **Rewards & Leaderboards** – Motivates eco-friendly actions.
- **Real-Time Collection** – Tracks and optimizes waste tasks.
- **Geo Reporting** – Reports dumping sites and shows facility locations.
- **In-App Training** – Guides users on segregation & composting.

DESIRED OUTCOMES:

- **Better Waste Tracking** – Record and monitor all waste.
- **Active Citizen Participation** – Encourage segregation and reporting.
- **Reduced Environmental Impact** – Minimize dumping and landfill load.
- **Informed Communities** – Provide training and facility access.
- **Efficient Collection** – Optimize routes and task management.

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TECHNICAL APPROACH

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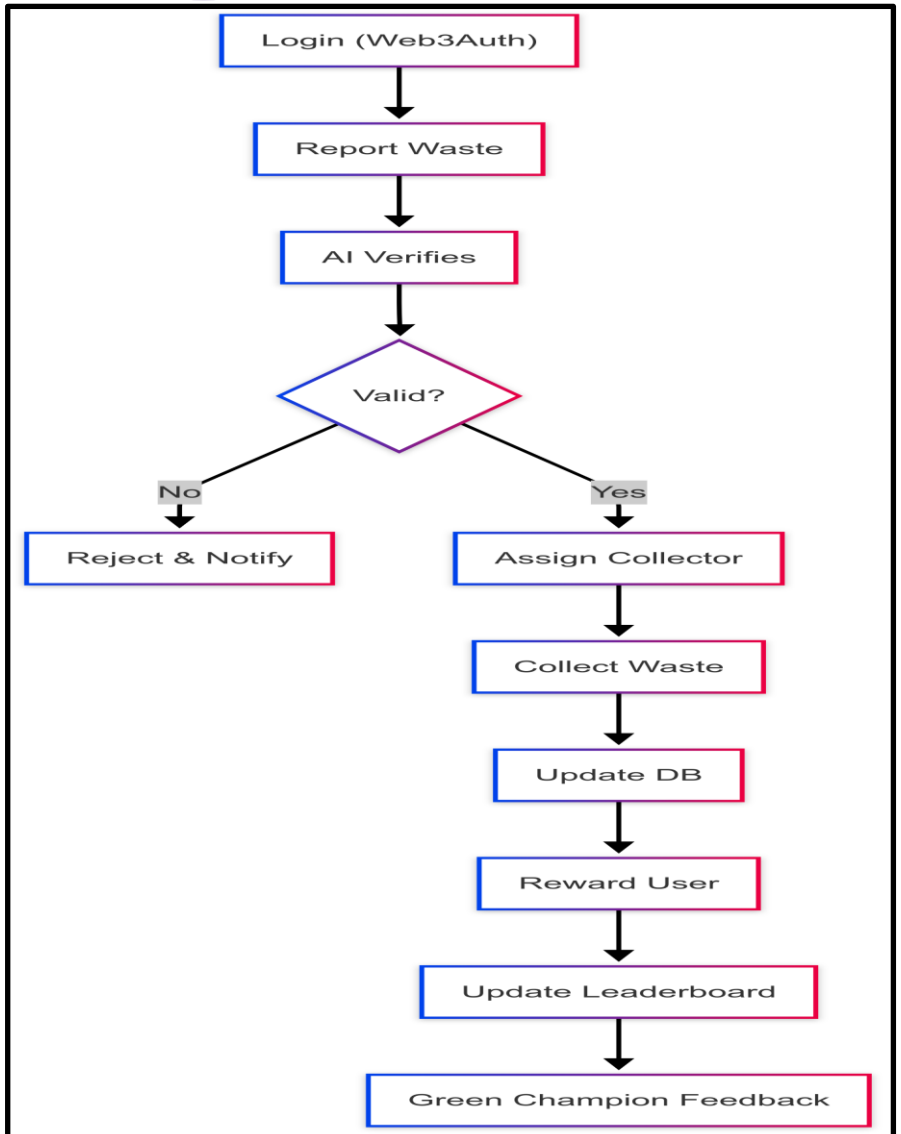
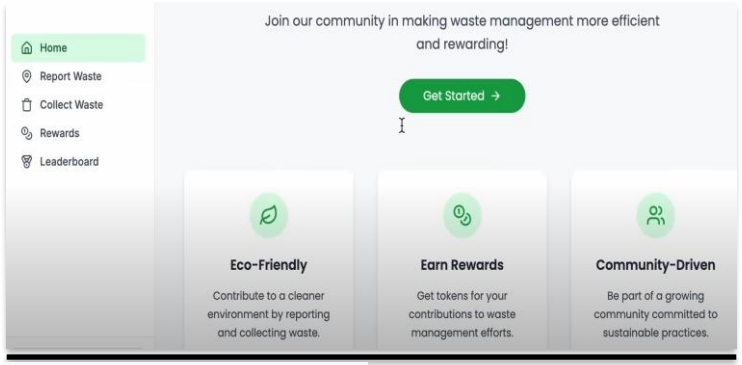
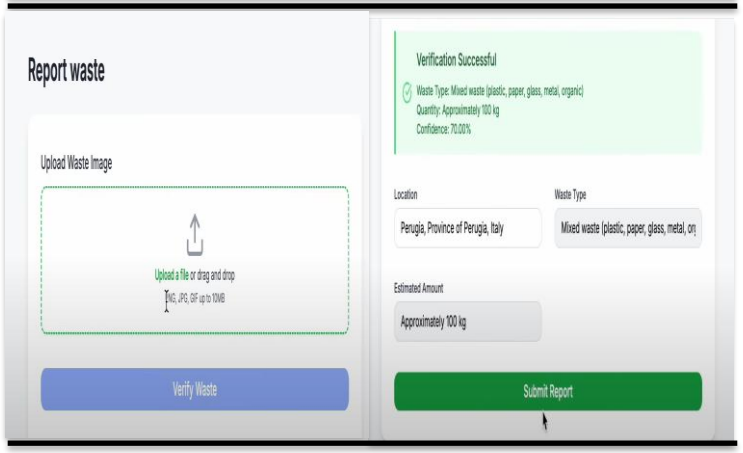
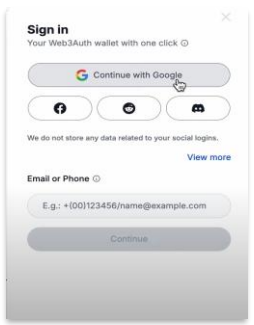


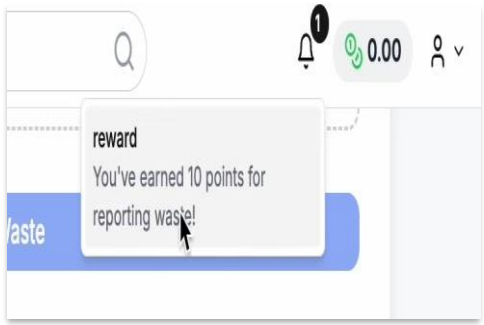
TECH STACK:

- **Frontend:** Next.js, TypeScript, Tailwind, Shadcn, Preline, Tabler Icons.
- **AI/ML:** Google Gemini AI, OpenAI (optional)
- **Auth:** Web3Auth
- **Database:** Neon DB, Drizzle ORM
- **Backend:** Next.js API
- **Deployment:** Vercel, ESLint, Prettier.
- **Utils:** Geo-tagging, React Hooks, Tailwind plugins.









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FEASIBILITY AND VIABILITY

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ANALYSIS & FEASIBILITY:

- **Technical Feasibility:** Built with Next.js 14 frontend, Drizzle ORM backend, Google Gemini AI for waste verification, and Web3Auth for secure authentication; supports real-time tracking and scalable cloud infrastructure.
- **Cross-Platform Compatibility:** Accessible via web and mobile, designed for responsive use across urban and semi-urban areas with multilingual support.
- **Scalability:** Cloud-based deployment allows scaling from small towns to large cities with minimal infrastructure changes.
- **AI/ML Integration:** Uses Google Gemini AI and OpenAI APIs for waste image verification and reporting accuracy; continuous training improves model reliability.
- **User Adoption:** Gamification, rewards, and simple UI promote engagement among citizens, waste workers, and local bodies.
- **Cost Considerations:** Moderate development costs with potential savings through improved waste segregation, reduced landfill load, and community participation incentives.

SWOT ANALYSIS:

- **Strengths:**
  - AI improves accuracy and boosts engagement.
  - Secure, scalable platform with Web3Auth.
- **Weaknesses:**
  - Relies on smartphones and internet access.
- **Opportunities:**
  - Growing government and NGO focus.
  - Expansion to recycling and smart cities.
- **Threats:**
  - Privacy, digital divide, and competition risks.

POTENTIAL CHALLENGES AND RISKS:

- **AI/ML Integration:** Edge cases in waste classification; needs ongoing training.
- **Scalability:** High costs with GPS/sensor rollout; phased approach needed.
- **User Adoption:** Maintaining engagement; incentives and penalties needed.

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IMPACT AND BENEFITS

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SOCIAL BENEFITS:

- Empowered citizens via AI-driven reporting and education.
- Gamification incentivizes eco-friendly actions.
- Safer waste workers with digital training and tools.

ECONOMIC BENEFITS:

- Jobs in AI monitoring, reporting, and recycling.
- Revenue from recycling, compost, and Waste-to-Energy**(WtE**).
- Reduced landfill management costs.

ENVIRONMENT BENEFITS:

- Cuts illegal dumping through real-time reporting.
- Boosts recycling, composting, and energy recovery.
- Lowers emissions and improves urban cleanliness.

IMPACTS:

- AI verification improves report accuracy.
- Rewards drive better segregation and participation.
- Enhanced tracking enables timely waste management.
- Data aids local bodies in decision-making.

USE CASE:

- Users earn rewards for consistent correct waste segregation, redeemable for bill discounts or benefits.
- AI flags incorrectly segregated waste and sends alerts to users for correction.
- Digital training modules educate citizens and waste workers on segregation and safety.
- Geo-tagged photos help authorities quickly spot and address illegal dumping.
- Community leaderboard tracks top contributors, encouraging competition and public participation.

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RESEARCH AND REFERENCES

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BOOK REFERENCE:

- **Solid Waste Management in Indian Cities** – *SpringerLink*
  - Covers policies, collection, and treatment systems
  - Link: https://link.springer.com
- **Waste-to-Energy: Technologies and Project Implementation** – *Elsevier*
  - Explains WtE plants, biogas, and recycling processes
  - Link: https://www.sciencedirect.com/book/9780128155547/substitute-natural-gas-from-waste

ONLINE RESOURCES:

- CPCB MSW Annual Report 2021–22
  - link: https://ceew.in/publications
- Swachh Bharat Mission Guidelines
  - link: https://swachhbharatmission.gov.in

RESEARCH WORK:

- AI-powered Waste Detection – Use of Gemini AI for real-time image classification and waste verification.
- Drizzle ORM & Neon DB used for lightweight and scalable backend data tracking.
- Web3Auth enables secure, decentralized citizen authentication.